FY-924

Reg. No. :

Name : ...

FIRST YEAR HIGHER SECONDARY IMPROVEMENT/SUPPLEMENTARY EXAMINATION, OCTOBER 2026

Part – III

Time : 2 Hours

PHYSICS

Cool-off time : 15 Minutes

Maximum : 60 Scores

General Instructions to Candidates :

- There is a 'Cool-off time' of 15 minutes in addition to the writing time.
- Use the 'Cool-off time' to get familiar with questions and to plan your answers.
- Read questions carefully before answering.
- Read the instructions carefully.
- Calculations, figures and graphs should be shown in the answer sheet itself.
- Malayalam version of the questions is also provided.
- Give equations wherever necessary.
- Electronic devices except non-programmable calculators are not allowed in the Examination Hall.

വിദ്യാർത്ഥികൾക്കുള്ള പൊതുനിർദ്ദേശങ്ങൾ :

- നിർദ്ദിഷ്ട സമയത്തിന് പുറമെ 15 മിനിറ്റ് 'കൂൾ ഓഫ് ടൈം' ഉണ്ടായിരിക്കും.
- ചോദ്യങ്ങൾ പരിചയപ്പെടാനും ഉത്തരങ്ങൾ ആസൂത്രണം ചെയ്യാനും 'കൂൾ ഓഫ് ടൈം' ഉപയോഗിക്കുക.
- ഉത്തരങ്ങൾ എഴുതുന്നതിനു മുമ്പ് ചോദ്യങ്ങൾ ശ്രദ്ധാപൂർവം വായിക്കണം.
- നിർദ്ദേശങ്ങൾ മുഴുവനും ശ്രദ്ധാപൂർവം വായിക്കണം.
- കണക്ക് കൂട്ടലുകൾ, ചിത്രങ്ങൾ, ഗ്രാഫുകൾ, എന്നിവ ഉത്തരപേപ്പറിൽ തന്നെ ഉണ്ടായിരിക്കണം.
- ചോദ്യങ്ങൾ മലയാളത്തിലും നല്കിയിട്ടുണ്ട്.
- ആവശ്യമുള്ള സ്ഥലത്ത് സമവാക്യങ്ങൾ കൊടുക്കണം.
- പ്രോഗ്രാമുകൾ ചെയ്യാനാകാത്ത കാൽക്കുലേറ്ററുകൾ ഒഴികെയുള്ള ഒരു ഇലക്ട്രോണിക് ഉപകരണവും പരീക്ഷാഹാളിൽ ഉപയോഗിക്കുവാൻ പാടില്ല.

SECTION – A

Answer any 5 questions from 1 to 7. Each question carries 1 score. **(5 × 1 = 5)**

1. Which of the following has least significant figures ?

   (a) 0.80760 (b) 0.80200

   (c) 0.08076 (d) 80.200

2. State whether the given statement is true or false :

   "The sign of acceleration alone does not tell about the speed of the particle, either it is increasing or decreasing."

3. If $|A + B| = |A - B|$, the angle between the vectors A and B are _____.

4. The average kinetic energy of gas molecule is proportional to

   (a) Potential energy (b) Temperature

   (c) Density (d) Size

5. Stress required to double the length of a wire of Young's modulus Y is

   (a) Y/2 (b) 2Y

   (c) Y (d) 4Y

6. Viscosity of gases _____ with temperature, whereas viscosity of liquids _____ with temperature. (Increase/Decrease).

7. A particle executing Simple Harmonic Motion is an example of

   (a) Acceleration of constant magnitude and direction

   (b) Acceleration of changing magnitude and direction

   (c) Acceleration of constant magnitude but changing direction

   (d) Acceleration of changing magnitude but constant direction

SECTION – B

Answer any 5 questions from 8 to 14. Each question carries 2 scores. **(5 × 2 = 10)**

8. Can a body have zero velocity, still accelerating ? Explain.

9. What is Poisson's ratio ? Explain its significance.

10. (a) Hydrostatic pressure is a _____ quantity. (Vector/Scalar) **(1)**

    (b) Explain why ? **(1)**

11. Triple point of carbon dioxide is –56.6 °C. Express this temperature on Fahrenheit scale.

12. State Kepler's law of areas. Write a mathematical expression of it.

13. A body which is initially at rest undergoes one-dimensional motion. Show that the power delivered to it at a time 't' seconds is proportional to time 't'.

14. Distinguish between transverse and longitudinal waves.

SECTION – C

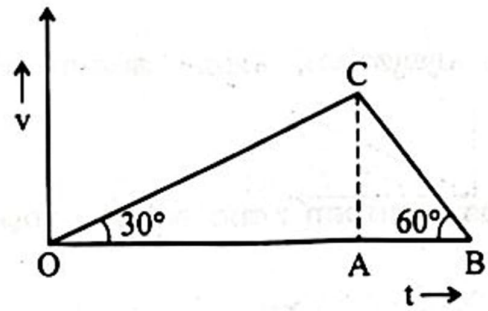
Answer any 6 questions from 15 to 21. Each question carries 3 scores. **(6 × 3 = 18)**

15. (a) Velocity and Electric current cannot be added together. Name the principle behind this. **(1)**

    (b) A boy recalls the relativistic variation of mass wrongly as $m = m_0 / \sqrt{(1 - v^2)}$, where '$m_0$' is the rest mass and 'v' is the velocity.

    Using dimensional method, put the missing term velocity of light 'c' in proper place and correct the equation. **(2)**

16. Study the velocity-time graph of a body given below and answer the following questions :



(a) Obtain the ratio of average acceleration during the intervals 'OA' and 'AB'. (2)

(b) The area of triangle OBC will represent which physical quantity ? (1)

17. Using parallelogram law of vector addition, find the magnitude and direction of the resultant of two vectors A and B in terms of their magnitudes and angle between them.

18. A man jumping out of a slow moving bus falls forward.

(a) This is due to _____. (1)

(b) Which Newton's law gives the above concept ? (1)

(c) State the law. (1)

19. What are the factors on which moment of inertia of a rigid body depends ?

20. (a) Name the four processes taking place in a Carnot's cycle. (2)

(b) Can a Carnot engine work if its sink and source are interchanged ? (1)

21. (a) What are sound beats ? (1)

(b) Two sitar strings A and B playing notes are slightly out of tune and produce beats of frequency 6 Hz. The tension in the string A is slightly reduced and the beat frequency is found to be 3 Hz. If the original frequency of A is 324 Hz, what is the frequency of B ? (2)

SECTION – D

Answer any 3 questions from 22 to 25. Each question carries 4 scores. **(3 × 4 = 12)**

22. A particle is projected with a velocity 'v_0' making an angle 'θ' with the horizontal. Draw the diagram and derive expressions for
    (i) Time of flight
    (ii) Maximum height attained

23. (a) State law of conservation of angular momentum. **(1)**
    (b) A boy stands at the centre of a turntable with his arms stretched out. The table rotates at a speed of 40 rev/min. What will be the angular speed of the boy if he folds his arms, thereby reducing his moment of inertia to 2/5 of his initial value ? **(3)**

24. (a) Derive an expression for time period of oscillation of a simple pendulum assuming that the pendulum is in simple harmonic motion. **(3)**
    (b) If this is kept on an elevator descending with an acceleration of 'g/3', find new period of oscillation. **(1)**

25. (a) Explain the anomalous behaviour of water at 0 °C and its environmental significance. **(2)**
    (b) Calculate the amount of heat energy required to convert 10 kg of water at 100 °C. Given latent heat L = 22.6×10^6 J/kg. **(2)**

SECTION – E

Answer any 3 questions from 26 to 29. Each question carries 5 scores. **(3 × 5 = 15)**

26. For a particle to leave from earth's gravitational field it should be projected with a minimum velocity.
    (a) Name the velocity. **(1)**
    (b) Obtain an expression for it. **(2)**
    (c) An elephant and an ant are to be projected into space. Whether the velocities required for doing so is same or different ? Justify. **(2)**

27. Banking of roads helps to increase the limit of maximum safe speed of a vehicle at a curve.

(a) What is banking of roads ? (1)

(b) Draw a schematic diagram of a vehicle on a banked road and mark various forces. (2)

(c) Obtain an expression for maximum safe speed of a vehicle on a banked road with frictional forces. (2)

28. (a) What is mechanical energy ? (1)

(b) Prove that total mechanical energy of a freely falling body is conserved at every point of its fall. (3)

(c) What is the nature of force so that mechanical energy is conserved ? (1)

29. (a) State and prove Bernoulli's theorem in fluid flow. (3)

(b) Explain the mechanism of lift of an aerofoil using this theorem. (2)
